SCORE: Synthesising Complete, Owned, Responsive Environments for Simulation

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Abstract

Games, film, robotics and embodied AI all require 3D environments that can be physically simulated, in which every object is a separate body with its own collision geometry, mass and joints under the gravity of the ground it stands on, and which remain editable afterwards and owned by whoever made them. World models generate video frames that no physics simulator can open, and one-shot 3D generators produce a single fused scene without separate bodies or clear usage rights. SCORE is a framework that synthesises such environments. It builds them as film and game studios build scenes in OpenUSD, the open scene description standard, with each scene composed inside its parent scene and each kind of work on it (layout, surfaces, physics, lighting, motion, effects, sound and people) held in a layer of its own that is checked and changed independently of the others. A creator fixes the design from the top down, first an environment concept that sets its visual style and physical setting, then a concept image for each scene. Open vision foundation models, procedural generation tools and coding agents produce every layer below, each output passing measured acceptance tests, called gates, before the next is made, and every model licenses its output for commercial use, so the creator remains the author. Every step is identified by a hash of its inputs, so an edit reruns only the steps that depend on it. The result is a single OpenUSD environment whose physics description a simulator reads as written and which Newton steps with the MuJoCo solver. We describe the framework and its method, with a first environment as the evidence, whose four demonstrations remain open until its paid production run, called a batch, has completed. Each character also receives a simulated body in Newton, driven by the segment rotations of its SOMA body model. We close with the route towards subject-specific musculoskeletal bodies, a data loop in which generated movement, retained only when it is dynamically consistent, trains controllers that actuate a body's muscles, and layered training sets with physics-derived ground truth, on which estimators can learn to infer muscle activations, joint moments and ground reaction forces from monocular video. Code and paper are at https://github.com/Babon-Innovations-b-v/score (MIT licence).

Gap: Figure 1, one environment, every layer. Environment 1 from the top down: its physics (the bodies it stands on and their gravity), a top scene on its site, a parent scene with its child scenes, and one scene opened into its layers side by side (base, settle, physics, light, motion, effects, sound, people). It waits on environment 1's batch.

1 Introduction

Simulated 3D environments serve games, film, robot learning and embodied AI [1]. Across these uses they share a common structure of separate objects with collision geometry, mass and joints, a ground surface with its gravity, light and sound emitted by identifiable sources, and scenes that can be reopened, modified and simulated again. Authoring such an environment by hand is expensive, and a generative system that takes over this work must leave the result in the hands of its author,

since in the Game Developers Conference's 2026 survey 52 percent of game workers considered generative AI to be harming the industry [2].

World models generate an environment as a sequence of video frames [3], [4] that no physics simulator can step forward in time. 3D environment generators [5], [6] produce geometry, mostly as a single fused asset from which separate objects have to be recovered afterwards [7], [8]. Both operate in a single pass, so a fault in the output cannot be traced to the step that caused it. Procedural content generation systems [9], [10] and coding agents, language models that write scenes as executable programs [11], [12], [13], [14], [15], [16], keep the environment explicit and inspectable, but they either reconstruct a scene that already exists or assemble it from a fixed asset library. The models they produce often float above their supports or come apart [17], such agents assess their own geometry unreliably [18], and on scenes reconstructed from video the strongest model scores 0.91 on static scenes but only 0.67 on dynamic ones [19]. Open vision foundation models [20], large models pretrained on broad image data and reused across tasks, now recover 3D shape [21], segmentation masks [22] and depth [23] from images, and a coding agent can call them as tools, yet none of them gives a consistent visual style across objects that different models produced [24], [25], and none gives physical properties.

The components needed for the physics already exist. OpenUSD [26], the scene description that Pixar developed for its film production, stores each department's work in its own layer file and composes scenes from smaller ones by reference, and its physics schema describes rigid bodies, colliders, mass and joints within the same stage, the composed scene. MuJoCo [27] is a widely used physics engine for articulated bodies in contact, and Newton [28], built on the Warp GPU programming framework [29], runs MuJoCo's solver on the GPU [30] and loads OpenUSD scenes directly. What is missing is a route from what a creator intends to a complete environment in this form.

SCORE is that framework. Its environments are complete, since every object, surface, light and sound a scene shows exists wherever the viewer looks. They are owned, because every model in the framework permits commercial use of its output, so the creator holds the full rights. They are responsive, because every object carries its collider, mass and material and every moving part its joints, so a simulator steps the environment as written. They are editable, since each scene is a stack of layers produced by steps that are each identified by a key derived from their inputs, and a change reruns only the steps it affects. An environment can range from a house on a plot of ground to a system of worlds that move one another by gravitation.

We contribute the SCORE framework (Section 2), which turns a creator's locked design into a single OpenUSD environment that is physically simulable and opens unchanged in a game engine, a robot simulator or a film renderer. Its method (Section 3) builds a scene from the top down and layer by layer, with each stage passing measured acceptance tests before the next begins. The evidence plan (Section 4) comprises four demonstrations on a first environment, left as gaps until its batch has run. Section 5 sets out where the framework goes next, towards simulated people whose bodies are true to the individual person.

2 The framework

An environment is stored as a single OpenUSD record. Each scene is one stage, a composed OpenUSD scene, that sits within its parent scene and holds its child scenes, each referenced at the room that the parent's plan, a dimensioned drawing in metres, assigns to it, so that opening a top

scene opens everything below it. Each scene's layers are stacked in a fixed order of precedence, from a slot reserved for the creator's own edit layer at the top, through people, sound, light, motion, effects, crops, machines, physics, settle and ground, down to the base layer laid out from the locked plan. Each object corresponds to one real-world object and carries its kind, transform, collider and, for each of its parts, a surface from the library. The environment's physical setting is decided in its environment concept (EC), the top-level design the creator locks, as a plot of ground with its gravity and the path of its sun, as a single world, or as a system of bodies that move one another by gravitation. A real celestial body takes its values from a cited table, and an invented one is specified by its radius, mass, orbit and spin. The physics layer writes a scene's rigid bodies, masses, joints and the gravity of the body it stands on in OpenUSD's physics schema, so that any reader of that schema obtains the same scene.

The creator works in the creator cockpit, a single authenticated web page in which he locks the design, purchases paid computation and reviews what the framework reports. The work on an environment is represented as a table of stages derived from its designs, and the table for environment 1 has 261 rows. A node is one stage applied to one scene (written `assembly@hangar` for the assembly stage of the hangar scene) or to the environment as a whole, and a single runner produces, evaluates and accepts every node. Once the creator has approved a design, it runs without further approval.

Keys. A node is computed only when its key has no completion record, as in a content-addressed build system. The key is a hash over the stage and scene, the version of its code, its contracts, parameters and random seed, the pinned container images of its models, every file it reads and the accepted output of every node it reads. The code version covers only code that can shape the output. Code that decides where and how a job runs changes what a run costs but never what it produces, and is therefore excluded from every key. An edit or a fixed fault changes only its own node's key and the keys of the nodes downstream of it, so only those steps run again. Outputs are stored under their content hash and never overwritten.

Gates and accepts. Each gate is an automated acceptance test that returns pass, fail or unknown, together with its measurements and a reason. Missing evidence counts as unknown, and unknown blocks acceptance just as fail does. The runner accepts an output automatically when every gate has passed and the output is no worse than the last accepted one, and downstream nodes read only accepted outputs. Four actions are reserved for the creator. He may reject an accepted output, which marks every node that read it as stale, accept a worse output with a stated reason, waive one gate for one key, and override the runner for one paid run. Coding agents are prevented from taking these actions by a store that refuses such records when they come from an agent's process, and by a hook in the agent's harness.

Batches. Paid computation is purchased as a batch, a production run with a single cost estimate. The creator's approval sets the batch's budget across all of its attempts, and spending guards stop the batch when that budget is reached. Within a batch every paid job runs once, and a failed output becomes a batch gap, which the completion gate lists. Every paid stage is priced from a unit cost measured on the cloud billing ledgers. A stage that has not yet been measured in a run has no price, and its estimate is shown separately as a guess.

3 Method: building an environment

3.1 From concept to plan

The environment concept. The creator first locks the EC, which fixes the environment's visual style and physical setting together with his reference images and the scenes it contains, in a short structured interview conducted with the vision-language model Gemini 3.1 Pro [31]. An environment plan then places the site of each top scene on that ground and computes the sky visible from it.

A concept for each scene, top down. A concept image is then locked for each scene, parents before their child scenes. Each candidate image is one call to the image generation model Nano Banana Pro [32], made on a concept review page where the creator marks regions and requests a new image. A child scene is drawn only once its parent has been locked and planned, from a crop of the parent's concept, so that it fits the parent it stands in.

Inventory and plan. Gemini 3.1 Pro lists each object of the locked concept as a row of the scene inventory, giving its kind, count, anchor and size in metres, and the segmentation model SAM 3 [22] localises each row with a bounding box. Nothing outside the inventory is generated or placed. Nano Banana Pro then draws the scene from directly above, and Gemini 3.1 Pro writes the plan in metres from that drawing, the concept and the inventory. The plan gives the outline, walls, openings, levels, stairs, roofs and terrain, each item with its orientation, people as the spots on which they stand, and the rooms of the child scenes. Free checks, which make no paid model call, test that every row is placed as many times as its count, inside the outline and clear of other floor items, that every door and standing spot can be reached on foot, and that the child scenes tile their parent. Any remaining violations are returned to the model as notes at most twice, and only what still fails is recorded as a gap. The creator's edits on the plan review page always take precedence and cause everything that follows to be recomputed.

Shapes first. The route follows one coarse-to-fine loop at every scale. A parent scene receives its concept, plan and a rough block model, and its plan is divided into the shapes of its child scenes. Once those shapes are fixed, all child scenes are produced in parallel, each repeating the loop. The stages `rough-model` and `shell-split` are implemented but have not yet run.

3.2 Layers from row kinds

A scene's layers are derived from its design alone. Each inventory row's kind, determined from the noun in the inventory reading by a single table of rules, states what its objects do, whether they emit, move, lie loose, glow, grow, act or sound. A scene receives an effects, motion, physics, crops or people layer only for rows that do so, and always a light and a sound layer. Each layer is generated from its rows by the rule of their kind, so a new environment needs no hand-written list of layers. The scene record holds what no generated object covers (building shells, stairs, ground and water, the distant backdrop, lamps and the neighbouring scenes its doorways lead to), and its gate requires the shell to lie within 5 cm of the plan and every doorway to open onto a floor.

Models. Objects follow the prop route first, which turns a single image of an object into a 3D model. Nano Banana Pro draws a close-up of the object from its crop of the concept and redraws it at most three times when the shape check rejects it. The image-to-3D model Pixal3D [21] then reconstructs the mesh, and a likeness gate compares its silhouettes from eight viewpoints with the close-up. Only an object whose checks fail is built instead by a procedural builder, a program that a coding agent writes, and a thin straight object (a pipe, post or beam) whose every close-up failed is generated in code by sweeping its cross-section along its length.

Surfaces. Everything a scene shows comes from the creator's asset library, score_lib, or is recorded as a written exception. A generated model contributes geometry only, and each of its parts takes one material from a library of procedural materials, written as ProcFunc functions [33] on Infinigen's shaders [9] and coloured from the scene's palette. The finish route labels each face of the model from free renders, matches each finish to a library material, bakes it into textures and checks the bake against the close-up, requiring each finish to lie within a colour difference of ΔE 32 of its colour there. A free library check traces every material, model, light, sound and character of an accepted assembly back to the library or to an exception.

Settle and physics. Loose objects are dropped onto the generated ground in a rigid-body simulation, the settle, and an object that would rotate by more than 15 degrees or drift by more than 30 cm is recorded as a resting fault and keeps its laid-out pose. The settle writes the pose in which each object comes to rest as a layer. The physics trial then steps the composed scene in Newton, using MuJoCo's solver with Newton's contact handling under the gravity of the body the scene stands on, and evaluates it with the gates `reacted`, `at rest`, `energy` (no increase in the energy of undriven bodies) and `hinge` (a hinged door leaf swings to its joint limit). The trial moved from Blender's Bullet engine [34] to Newton on 27 October 2026. On a test scene at lunar gravity (two dropped crates, a driven door with a wheel fixed to it and a pushed stool), both engines gave the same result on every gate, and Newton brought the crates to rest on their targets where Bullet left them standing off by its contact margin of 8 mm. Newton's deterministic mode, which reproduced bitwise-identical results over six runs of that test scene, cannot handle scenes with many loose bodies, because its buffers exceed the 32-bit limit, so the trial and the settle run without it. Two runs of a 25-body test scene then agree within 5e-7 m for every body except one that falls off the edge of the floor. The settle moves to Newton in the same way, so that one engine steps both what comes to rest and what is tested, and one effect is simulated in Newton as a first case for the effects layer.

Light, motion, effects and sound. The sky is computed from the physical setting in the EC, never drawn, with each celestial body at its true direction, apparent size and phase as seen from the scene's site. The lamps are the inventory's glowing rows. One render per scene places each lamp in its own light group, and a free sweep mixes their intensities to bring the scene's brightness within a factor of 1.5 of its concept's. Every object of a moving row moves by the rule of its kind at a speed taken from cited real machines, and is gated on nothing coming within 3 cm of anything else. Emitters are measured against real footage. Each sound source receives a library file or one take generated by the text-to-audio model MOSS-SoundEffect [35], gated on an audio-text similarity score from CLAP [36] of at least 0.30.

People. People are models like any other, each made from a close-up per person row, reconstructed by Pixal3D and given its materials by the finish route. The SOMA parametric body model [37], estimated from the close-up by SAM 3D Body [38], is fitted beneath the model as its skeletal rig and animated with motion clips from Kimodo [39]. A shell gate accepts a clothed appearance when 99 percent of the points of its non-rigid parts lie within 5 cm of the body. Each person stands on a person spot of its scene's plan.

Sim bodies. Each character also receives a sim body, its physical twin in Newton, so that the people in an environment are physically simulated as well as animated. The minimal body is an articulated rigid-body model built directly from SOMA's 23 segments, with a ball joint at each SOMA joint frame. It is driven by the character's SOMA segment rotations without scaling or per-person setup, its mass is distributed according to its segment volumes, and it stands on a SCORE floor so that its ground reaction forces arise from contact. Muscles are added as a minimal first step. Built this way, the body follows SOMA's rotations to within 6.9e-9 m, and in Newton it matches

plain MuJoCo to within 3.4e-8 after 200 steps.

Gap: Newton for the settle, one effect and the sim body. The settle and the physics trial on one engine, with the resting and physics gates per scene against the Bullet settle; the one effect stepped on Newton; and a sim body walking on a SCORE floor, its joint angles and ground forces against OpenCap's LabValidation motion capture and force plates [40]. It waits on these three stages being built and run on a cloud card.

Completion. A scene is complete when its completion gate passes on its final assembly, with every inventory row made or dropped with a reason, nothing left unmade, no resting fault, and every layer and every visible scene present. A fault the creator chooses to ship with is an accepted gap, recorded with who decided, when and why. The gate lists every gap by name and is never relaxed. A hook prevents a coding agent from claiming that a scene is done while its gate fails, because agents judging their own scenes agree with a measured geometry score only 45.8 percent of the time [18]. The films of a complete scene are rendered from scripts the creator has locked.

4 Evidence: environment 1

Environment 1 serves as the evidence. The creator did not accept its first run, and its store of outputs was lost on 19 October 2026, so this draft shows none of its outputs, and each demonstration below will be produced by its new batch. Figure 1 is the first.

Gap: Demonstration 2, physics that holds. Environment 1's scenes stepped on Newton under their own body's gravity: loose objects come to rest, doors swing on their hinges, with the physics trial's gates per scene and Bullet run on the same scenes as the baseline. It waits on environment 1's batch and on Newton taking over the settle.

Gap: Demonstration 3, one environment, many uses. The same OpenUSD environment opened unchanged in a game engine, in a robot simulator and in a film render, with what each reads of its layers and what it drops. It waits on environment 1's batch and on the engine adapter.

Gap: Demonstration 4, an edit that stays cheap. The creator changes one thing in a locked scene, and only the nodes whose keys it moves run again; the nodes rerun, time and euros against a full rebuild. It waits on environment 1's batch and on the edit layer being carried from one assembly to the next.

Gap: Table 1. Per scene of environment 1: inventory rows, models generated and code-built, layers, gaps before and after the batch, time, euros and the creator's verdict. It waits on environment 1's batch and the creator's review.

Disclosure. The creator's locks, picks, accepts and waivers are his own actions in the cockpit, and everything between them is automatic. Coding agents wrote most of the framework's code under the author's direction, and 2,413 of its 2,436 commits list a coding agent as co-author. Manual fixes to a scene are recorded as the creator's edits.

5 Future work: people true to the person

The minimal sim body is a starting point, and what follows describes goals rather than results.

Bodies true to the person. The sim body is to become a full musculoskeletal model that is specific to the individual, older people and people with a curved spine included, rather than a scaled generic template. Joint centres are to be derived from bony landmarks on the SOMA mesh,

followed by segment masses, muscles, joint ranges of motion, the spine and bone geometry, with each input that SOMA does not provide marked as assumed and given a plausible range. A full-body musculoskeletal model with 416 muscles already runs in Newton but still drifts 1.5 cm from plain MuJoCo after 300 steps, so Newton is first to be validated for such bodies. The body's joint angles, and then its net joint moments and ground reaction forces, are to be validated against OpenCap's LabValidation data.

Scenes from motion. Motion is to come from the SOMA ecosystem, either estimated from video with GEM-X [41] or generated from text and constraints with Kimodo [39], in practically unlimited quantity. The terrain a motion implies becomes a SCORE scene, so the movement is kept in its setting without the source video or any information that identifies a person. The sim body then follows each movement as a prescribed trajectory on the scene's ground, and the simulation solves for the ground reaction forces from contact, the net joint moments, and the activations and forces of a 416-muscle model that reproduce it. Any force and moment that neither the ground contact nor the muscles can supply remains as the residual at the pelvis. It is recorded for every frame and never applied to the body, so it quantifies how dynamically consistent the movement is, and only movements with a small residual are kept. Biomechanics reduces and reports the same residual to assess recorded motion [42], [43], and PhysibleMDM [44] minimises an inverse-dynamics residual to make generated motion feasible for a torque-driven robot; we apply the criterion to generated motion in a muscle-driven body.

Layered training sets. The same simulations yield layered training sets. Each sample pairs a rendered image of a clothed person in a complete scene with physics-derived ground truth for every layer beneath it, namely the SOMA body and its joint angles, the ground reaction force under each foot from simulated contact, the muscle activations and forces, the residual, and the depth, segmentation and layout of the scene. Their main purpose is to train estimators that infer muscle activations, joint moments and ground reaction forces directly from monocular video, quickly enough for routine clinical use. Recorded data cannot provide this supervision at scale, because muscle forces cannot be measured non-invasively and force plates and electromyography confine measurement to a laboratory, whereas the simulation can generate the bodies, movements and settings an estimator has to cover, including those of a specific population, so that domain-specific estimators can be trained for it. BioHuman10M [45] already pairs video, part of it rendered, with the activations of a 112-muscle model and with ground reaction forces predicted by a learned model from recorded motion. Our sets would differ in scenes generated for the purpose under varied lighting, terrain and clothing, in ground reaction forces from contact with each scene's own ground, in a 416-muscle model and in a residual for every sample, and their value depends on whether estimators trained on them transfer to real force-plate and electromyography recordings.

Brains. A brain is a trained controller that actuates a sim body's muscles, and its role is to extend the training data beyond what a motion generator produces. Trained on the retained movements together with the muscle forces that produce them, it is placed in SCORE's synthesised worlds under their own gravity and terrain, where it is to adapt to structures no recording contains, and each run it drives adds movement to the training sets. Muscle-driven controllers such as KINESIS [46] and MuscleMimic [47] are trained on motion capture, as are torque-driven ones such as PHC [48]. To our knowledge, no muscle-driven controller has yet been trained on generated motion selected by its residual, and the claim holds only once such a controller is compared with one trained on motion capture and with one trained on the same generated motion unfiltered.

The data loop. The amount of data the loop produces scales with the compute put behind it, since every scene, movement and run adds samples. This is why SCORE is open, so that others

can apply their own compute to the same framework and share the resulting datasets. Estimators specialised for clinical use would be trained outside SCORE, on data that never enters it.

6 Limitations and conclusion

A single creator judges the results, and he has not yet accepted a run. Closed models (Nano Banana Pro, Gemini 3.1 Pro) draw the concepts and close-ups and read and plan every scene, and may change or be withdrawn. The design step, the layers derived from row kinds, the finish route, the shapes-first loop and the scene tree have not yet carried a scene through a batch. The edit layer is so far only a reserved slot. The settle, the effects and the sim body do not yet run in Newton as stages, and no game engine or robot simulator has yet opened the result. Next come environment 1's batch with its four demonstrations, an engine adapter, the edit layer carried across assemblies, further environments, open models replacing the closed ones, and the data loop, in which residual-filtered generated movement trains muscle-driven controllers whose value a comparison with motion capture has yet to show. SCORE synthesises an environment as studios build one, scene by scene and layer by layer, in a form a simulator steps as written, and an accepted environment will show whether the approach holds.

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